A Domain Language for Specifying Controlled Methods

A controlled-method specification pitch for reproducible research teams

Why methods drift

Prose is necessary but not sufficient

- A method description can be readable while leaving stages, units, and acceptance predicates ambiguous.
- The remedy is a small typed contract that can be validated before compilation.

The controlled pipeline

- Parse the method specification, validate staged dependencies, compile deterministic outputs, and emit a receipt.
- Exact-value tests and negative controls protect the surfaces most likely to drift.

Source: projects/templates/template_methods_paper/src

What remains bounded

- A passing compiler proves local method consistency, not scientific truth or external validity.
- Forks must add domain evidence and publication authorization before extending the claim.

Source: projects/templates/template_methods_paper/README.md

Adoption path

- Choose one existing workflow, encode its invariant stages, and compare the receipt across two deterministic runs.
- Use the failed case as the design signal; do not hide it behind a skipped tool.

Source: projects/templates/template_methods_paper/tests